KONE		TITLE: **LCEKDM Diagnostic Code List**	DRAWING NO: **994525D02**
COMPILED BY: **HAT/A. Calcagno, R. Jokinen, J. Laaksonheimo, E. Putkinen, J. Nikander, A. Pikivirta**		PRODUCT: **KDM**	ISSUE: **V.2**
CHANGED BY:		CHECKED BY: **HAT/Risto Jokinen**	LANGUAGE: **en**
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1000 - 1999 are faults that prevent normal operation even if fault is cleared
2000 - 2999 are faults that prevent or stop running as long as fault is active
3000 - 3999 are warnings that do not prevent elevator operation
6000 - 6999 are for info
(A) after subcode = supervision can be adjusted with parameters
Data 2 mode information: 1=normal, 2=inspection, 3=correction, 4=relevelling, 5=reduced speed

Main code	Sub code	Data 1	Data 2	Name	Description	Reason	Action	Detection	Operation	Recovery	Testing
102 Drive electronics											
	1024	0	0	Motorcurrent switch off fault	Motor current detected although motor bridge IGBTs are switched off.	Broken motor bridge IGBTs. Broken current sensors. Disturbance in current measurement. See also 130_5000_219.	Check fault 130_5000_219. Cycle power. If problem persists then replace drive.	Motor current is measured by current sensors.	Drive is faulted.	By user: switch to inspection/RDF mode or cycle power.	-
104 Motor protection											
	1035 A	0	0	Motor overload counter fault	Motor overload protection tripped too often.	Fault 2005 indicated too often.	Check motor overload limits. - Motor overload full speed current limit (6_68) - Motor overload full speed current limit (6_69) Check motor setup type (6_60). Check machinery parameters (6_80-6_94). Check elevator parameters (6_1-6_11). Check elevator mechanics. Check balancing.	This fault occurs if there's too many consecutive 104_2005 faults.	Drive is faulted.	By user: switch to inspection/RDF mode or cycle power.	-
	2004 A	0	0	Motor temperature fault	Motor thermistor or NTC-sensor indicates overtemperature.	Motor overheated. Motor thermistor PTC or NTC broken. Motor thermistor PTC or NTC cable disconnected. Wrong motor overtemperature limit (6_67). Only for NTC. Disconnection supervision could be unreliable in very cold temperature.	Check motor cooling. Check motor thermistor and thermistor wiring. Check motor overtemperature limit (6_67). Check motor temperature from RT display (6_75_45, shows temperature in Celsius with NTC or --- with PTC)	Drive measures voltage over thermistor or temperature with NTC. With NTC supervision limit can be adjusted with parameter 6_67.	Drive is faulted.	Automatically after about 5 mins (or by switching to inspection/RDF mode or cycling power) when sensor indicates normal temperature.	Disconnect thermistor/NTC cable and/or replace thermistor/NTC with potentiometer and adjust resistance to trip supervision.
	2005 A	0	0	Motor overload fault	Measured motor load too high.	Wrong motor overload limits. - Motor overload full speed current limit (6_68) - Motor overload full speed current limit (6_69) Mechanical problem in elevator. Wrong parameters.	Check motor overload limits. - Motor overload full speed current limit (6_68) - Motor overload full speed current limit (6_69) Check motor setup type (6_60). Check machinery parameters (6_80-6_94). Check elevator parameters (6_1-6_11). Check elevator mechanics. Check balancing.	Drive measures motor current during acceleration and full speed. If currents are greater than set limits for more than 2 seconds, supervision is tripped. Limits can be adjusted with parameters 6_68 and 6_69.	Stops immediately.	Automatically	Measure current needed by elevator and set limits lower than measured value.
	3044 A	0	0	Motor overload warning	Measured motor load too high.	Wrong motor overload limits. - Motor overload full speed current limit (6_68) - Motor overload full speed current limit (6_69) Mechanical problem in elevator. Wrong parameters.	Check motor overload limits. - Motor overload full speed current limit (6_68) - Motor overload full speed current limit (6_69) Check motor setup type (6_60). Check machinery parameters (6_80-6_94). Check elevator parameters (6_1-6_11). Check elevator mechanics. Check balancing.	Drive measures motor current during acceleration and full speed. If currents are greater than set limits for more than 1 second, supervision is tripped and warning is displayed. Limits can be adjusted with parameters 6_68 and 6_69.	Warning is displayed.	Automatically	Measure current needed by elevator and set limits lower than measured value.
	3057 A	0	0	Motor temperature warning	Motor thermistor or NTC-sensor indicates overtemperature.	Motor overheated. Motor thermistor NTC broken. Motor thermistor NTC cable disconnected. Wrong motor overtemperature limit (6_67). Disconnection supervision could be unreliable in very cold temperature.	Check motor cooling. Check motor thermistor and thermistor wiring. Check motor overtemperature limit (6_67).	Drive measures voltage over thermistor or temperature with NTC. With NTC supervision limit can be adjusted with parameter 6_67.	Warning is displayed.	Automatically after about 5 mins (or by switching to inspection/RDF mode or cycling power) when sensor indicates normal temperature.	Disconnect thermistor/NTC cable and/or replace thermistor/NTC with potentiometer and adjust resistance to trip supervision.
105 Power supply											
	2007	0	0	Power supply	DCBG-CPU 24V out of operational limits	Too high or low 24V supply voltage (<16V or >35V). KDM Drive internal 24V power supply broken. Broken DCBG-CPU.	Check cable from DCBG XAP1 to 367 I/O Board X7. Check 367 I/O board X85 connector. Change 367 IO board. Change KDM Drive. Change DCBG board.	DCBG-CPU has software supervision for 24V.	Stops immediately	Automatically	Lower 24V supply under 16V during boot or under 14V while running.
Main code	**Sub code**	**Data 1**	**Data 2**	**Name**	**Description**	**Reason**	**Action**	**Detection**	**Operation**	**Recovery**	**Testing**
106 Self-test											
	1001 A	0	0	Speed supervision counter	Tacho fault counter has exceeded the limit set in parameter 6_31.	Too many 2009, 2013, 5000_41 or 5000_48 faults. Counter value is adjustable by parameter 6_31. Default value is 3.	Check parameters. Check encoder fixing. Check encoder wire shielding. Check balancing.	Tacho fault counter is increased when run has been stopped by any one of the speed supervisions: 108_2009, 108_2013, 130_5000_41 or 130_5000_48. Counter is resetted after successful run.	Drive is faulted.	By user: switch to inspection/RDF mode or cycle power.	Disconnect motor encoder from drive and try to run several times.
	1044	0	0	Motor torque supervision counter	Motor torque is too high while moving.	Too many 2069 faults.	Check brakes. Check LWD. Check drive parameters: - Elevator nominal load (6_4) - Balancing (6_10) - KTW/Q factor (6_25) - rope weight (6_26)	Motor torque exceeds 80% of nominal torque for longer than 2 seconds while car is moving.	Stops immediately	Automatically	
	1056	0	0	Movement risk supervision counter (motor movement)	Sequence control has detected failure which could cause unintended movement	Fault 2110 indicated too often.	Check stop delay parameter (6_43). Check brakes according to AS instruction. Check manual brake opening device.	Motor movement is monitored from encoder at each start\stop.	Drive is faulted.	By user. Switch to inspection/RDF mode or cycle power.	Using modified version of BCB25\BCX08 or by causing movement during start or stop sequence.
	1064	0	0	Movement risk during start or stop sequence counter (HW failure)	Sequence control has detected failure which could cause unintended movement	Fault 2111 indicated too often.	Check brake supply wiring and all module groundings and shieldings. Replace brake controller.	With BCB25 brake current is monitored during start/stop sequence. With BCX08 brake relay fault is monitored during the whole run.	Drive is faulted.	By user. Switch to inspection/RDF mode or cycle power.	Using modified version of BCB25\BCX08.
	2110	0	0	Movement risk supervision (motor movement)	Sequence control has detected failure which could cause unintended movement	Problem with machinery brakes. Wrong brake adjustment. Manual brake opening device stuck. Too weak brakes. Brake close time is longer than brake stop delay parameter (6_43).	Check stop delay parameter (6_43). Check brakes according to AS instruction. Check manual brake opening device.	Motor movement is monitored from encoder at each start\stop. If brake adjustment is not correct, brakes are too weak or stop delay parameter 6_43 has been set too low, movement is detected during start and stop sequence and drive is faulted	Drive is faulted.	Automatically	Using modified version of brake controller or by causing movement during start or stop sequence.

	2111	0	0	Movement risk during start or stop sequence (HW failure)	Sequence control has detected failure which could cause unintended movement	Problem with drive brake controller. Start/stop control has detected unexpected brake current which could cause unintended movement. Wiring mistake.	Check brake supply wiring and all module groundings and shieldings. Replace brake controller	Brake current is monitored during start/stop sequence. With BCX07 brake relay fault is monitored during the whole run.	Drive is faulted.	Automatically	Using modified version of brake controller.
107 Load weighing											
	2008	0	0	Load weighing	Load weighing device (LWD) fault	Broken sensor. Problem with sensor cabling. Relevelling tried without LWD setup.	Check LWD sensor. Check LWD wiring. Make LWD setup.	Drive measures sensor current and if value is too low or high supervision is tripped. Levelling is not allowed without LWD setup.	Drive will not start.	Automatically when signal is in range or LWD setup is done.	Disconnect LWD from drive.
	3020	0	0	Load weighing warning.	Load weighing warning	Bad/misplaced LWD sensor. Problem with sensor cabling. LWD setup not done.	Make LWD setup. Check LWD wiring. Check LWD sensor.	Drive calculates load based on given setup and if value is too low or high warning is given.	Normal operation is allowed but warning is displayed after run.	Automatically when calculated load is in range or LWD setup.	Make LWD setup with wrong values.
108 Movement feedback											
	2009	position in shaft [cm]	speed+mode [cm/s] + last number is mode	Speed difference (reference vs. encoder)	Difference limit between speed reference and measured speed exceeded.	Wrong system parameterization. Problem with motor speed feedback. High friction affecting movement. Problem in brake or brake control. Wrong balancing. Safety chain opens during run.	Check parameters. Check encoder fixing. Check encoder wire shielding. Check balancing. Read shaft position (Data 1): Check it for safety devices or mechanical problem.	Drive compares speed reference and motor speed feedback and if difference is higher than 0.22 m/s supervision is tripped.	Stops immediately.	Automatically	Decrease speed controller's proportional gain 6_1 and run elevator.
	2012	position in shaft [cm]	speed+mode [cm/s] + last number is mode	Speed feedback saturated fault	Speed returned by primary speed feedback not chaning even car should move.	Primary encoder not connected or is broken.	Connect primary encoder. Replace primary encoder.	Primary encoder measured speed stays below 0.5mm/s for longer than 1,5 seconds even drive is at moving state.	Stops immediately or ramp stop to next floor. Drive faulted after predefined retries (6_31).	Automatically	
	2013	position in shaft [cm]	speed+mode [cm/s] + last number is mode	Dual speed supervision (encoder vs. motor frequency)	Difference limit between estimated speed and measured speed exceeded.	Wrong system parameterization. Problem with motor speed feedback. Safety chain opens during run.	Check parameters. Check encoder fixing. Check encoder wire shielding. Read shaft position (Data 1): Check it for safety devices or mechanical problem. Compare RT display values (6_75_1 vs. 6_75_34) should be close to each other.	Drive compares speed feedback signals from motor sensors and if the difference is higher than 0.22 m/s between them supervision is tripped. Tacho fault counter is increased except in inspection/rescue drive modes.	Stops immediately or ramp stop to next floor. Drive faulted after predefined retries (6_31).	Automatically	Connect external encoder to drive and try to run with RDF.
	2069	0	0	Motor torque supervision	Motor torque is too high while moving limit exceeded 3 times.	Motor torque exceeds 80% of nominal torque while car is moving.	Check brakes. Check LWD. Check drive parameters: - Elevator nominal load (6_4) - Balancing (6_10) - KTW/Q factor (6_25) - rope weight (6_26)	Motor torque exceeds 80% of nominal torque while car is moving.	Stops immediately.	Automatically	
	3084	0	0	Motor torque supervision	Motor torque is too high while moving.	Motor torque exceeds 80% of nominal torque while car is moving.	Check brakes. Check LWD. Check drive parameters: - Elevator nominal load (6_4) - Balancing (6_10) - KTW/Q factor (6_25) - rope weight (6_26)	Motor torque exceeds 80% of nominal torque while car is moving.	Stops immediately. Drive faulted after 3 times faulted in row.	Automatically	
109 Shaft signals											
	3049	0	0	77U or 77N position warning.	Slowdown switch check	77U or 77N is not active when deceleration is started to terminal floor from nominal speed.	Warning is given.	Switch state check from nominal speed --> jerk3	Warning only.	Automatically when problem is fixed and new setup is made	Set 77U/N too near of terminal floor, run setup and run to terminal floor.
	3065	floor	0	Car position 77 error	Drive has lost accurate position and needs a synchronisation run.	Problem with 77:U or 77:N signals Ropes slip too much.	Check 77:U and 77:N switches and magnets. Check balancing. Check shaft mechanics. Check RT display for elevator position drift (6_75_9).	Drive compares current and stored 77 positions and if difference is too big supervision is tripped.	Ramp stop to next floor.	Automatically after position information is consistent and position is verified.	Set 77U/N on at the middle of shaft.
	3066	floor	0	Car position 61 error	Drive has lost accurate position and needs a synchronisation run.	Problem with 61:U or 61:N signals Ropes slip too much.	Check 61:U and 61:N switches and magnets. Check balancing. Check shaft mechanics. Check RT display for elevator position drift (6_75_9).	Drive compares current and stored 61 positions and if difference is too big supervision is tripped.	Ramp stop to next floor.	Automatically after position information is consistent and position is verified.	Set 61U/N on at the middle of floors.
110 Drive temperature											
	2014	0	0	Heatsink temperature fault	Heatsink too hot (about 90C)	Heatsink temperature over 75C at start or over 90C during run. Heavy traffic. Drive fan is not operating. Dirty or dusty heatsink. Shaft or machine room temperature is too high.	Check drive fan (KR6 or KR7) or INU and AFE fan (KR8) operation. Check drive heatsink (KR7 or KR7) or INU and AFE heatsink (KR8) airflow. Check shaft or machine room temperature. Check RT display for drive heatsink temperature (6_75_40).	Drive measures heatsink temperature.	Drive is faulted at floor.	Automatically when heatsink temperature is low enough.	Heat heatsink.
	3010	0	0	Heatsink temperature warning	Heatsink too hot (about 75C)	Heatsink temperature over 75C when in inspection/RDF. Drive fan is not operating. Dirty or dusty heatsink. Shaft or machine room temperature is too high.	Check drive fan (KR6 or KR7) or INU and AFE fan (KR8) operation. Check drive heatsink (KR7 or KR7) or INU and AFE heatsink (KR8) airflow. Check shaft or machine room temperature. Check RT display for drive heatsink temperature (6_75_40).	Drive measures heatsink temperature.	Only inspection run is possible.	Automatically when heatsink temperature is low enough.	Heat heatsink.

Main code	Sub code	Data 1	Data 2	Name	Description	Reason	Action	Detection	Operation	Recovery	Testing
118 Shaft setup											
	1043	0	0	Shaft setup missing	No valid shaft setup.	No valid shaft setup. Shaft setup is not done or existing setup has been deleted.	Make shaft setup according to AM/AS instruction.	Drive monitors setup data and if data is invalid supervision is tripped.	Drive is faulted. Only setup or RDF/inspection drive is allowed.	By user: switch to inspection/RDF mode.	Delete existing setup by changing traction sheave diameter and switch to automatic use.
	2021	0	0	Setup start position fault	Setup started too close to 61:N vane or 61:U missing.	Setup started too close to 61:N vane or 61:U missing. Car is not well below 61:N edge before start or problem with 61 signals.	Start shaft setup below bottom floor (61:U and 30).	Drive monitors 61 signals during setup and if 61:N is active or 61:U is missing at start supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Start setup very near of 61:N edge.
	2022	0	0	77:S sequence fault	77:S edge detected in wrong place. 77:S must be inside 77:U/N zones.	Incorrect 77:S sequence. Correct sequence is: 77:S must be on at bottom floor 77:S must be off in middle floors 77:S must be on at top floor	Install 77:S according to vane diagram.	Drive monitors 77:U/N/S signals during setup and if sequence is invalid supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Cause wrong 77:S signal during setup.
	2023	0	0	77:U/N sequence fault	77:U/N edge detected in wrong place. N must be at bottom, U at top, U and N cannot be active at the same time and 77:S must be inside 77:U/N zones.	Incorrect 77:U/N sequence. Correct sequence is: 77:N must be on at bottom floor 77:N and 77:U must be off in middle real or dummy floor 77:U must be on at top floor	Install 77:U and 77:N according to vane diagram.	Drive monitors 77:U/N/S signals during setup and if sequence is invalid supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Cause wrong 77:U/N signal during setup.
	2025	0	0	61:N sequence fault	61:N edge detected in wrong place. 61:N must come on when 61:U is on and must go off when 61:U is off.	Incorrect 61:N sequence. Correct sequence is: 61:U active at floor and below floor 61:U and 61:N active at floor 61:N active at floor and above floor	Adjust 61:U and 61:N reader according to AM instructions. Install 61 magnets according to vane diagram.	Drive monitors 61:U/N signals during setup and if sequence is invalid supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Disturb 61 signals during setup
	2026	0	0	61:U sequence fault	61:U edge detected in wrong place. 61:U must come on when 61:N is off and must go off when 61:N is on.	Incorrect 61:U sequence. Correct sequence is: 61:U active at floor and below floor 61:U and 61:N active at floor 61:N active at floor and above floor	Adjust 61:U and 61:N reader according to AM instructions. Install 61 magnets according to vane diagram.	Drive monitors 61:U/N signals during setup and if sequence is invalid supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Disturb 61 signals during setup

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	2027	0	0	Maximum floor count exceeded	Maximum floor count exceeded. (max. 128 floors)	Maximum floor count exceeded. (max. 50 floors)	Install position magnets according to AM instructions Check positioning system.	Drive records floor information during setup and if maximum floor count is exceeded supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Make shaft setup with too many floors. (simulator)
	2029	0	0	Position calculation fault	Position is not increasing normally.	Calculated position is below bottom floor.	Check motor parameters (6_60 or 6_80 - 6_91). Verify RDF operation upwards and downwards. Restart shaft setup.	Drive calculates position from motor encoder during setup and if position is not increasing normally supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	
	2073	0	0	77:U/N NTS sequence fault	Wrong NTS switch order.	Incorrect 77:U/N sequence. Correct sequence is: 77:N must be on at bottom floor 77:N and 77:U must be off in middle real or dummy floor 77:U must be on at top floor	Install 77:U and 77:N according to vane diagram.	Drive supervises NTS switches during setup and if the switches are detected in wrong order supervision is tripped. State of NTS switches can be monitored using real-time display 6_75, signal 8.	Stops immediately		Swap NTS switches.
	6010	floor	overlap [mm]	Shaft setup minimum 61:U and 61:N overlap info	Reports minimum 61:U/N overlap	Reports minimum 61:U/N overlap when below 5 mm at Data 1 floor	If minimum overlap < 5 mm Adjust 61:U and 61:N reader according to AM instructions. Start new shaft setup.	Drive measures accurately every edge of 61 signals and calculates minimum overlap.	Value is shown after setup	-	Cause very short and very long overlaps
	6011	floor	overlap [mm]	Shaft setup maximum 61:U and 61:N overlap info	Reports maximum 61:U/N overlap	Reports maximum 61:U/N overlap when above 25 mm at Data 1 floor	If maximum overlap > 25 mm Adjust 61:U and 61:N reader according to AM instructions. Start new shaft setup.	Drive measures accurately every edge of 61 signals and calculates maximum overlap.	Value is shown after setup	-	Cause very short and very long overlaps
120 Start sequence											
	1037	0	0	Motorbridge fault	Motor bridge does not start.	Motorbridge does not start. Motorbridge broken.	Cycle power. If problem persists then replace drive.	Drive tries 10 times to start and if motor bridge does not start supervision is tripped.	Drive is faulted.	By user: switch to inspection/RDF mode or cycle power.	-
	1038	0	0	Main contactor control fault	Main contactor status indicates that contactor is not closing.	Main contactor or auxiliary contact broken. Problem with status wiring.	Check wiring. Check main contactor. Check auxiliary contactor block.	Drive monitors main contactor status input by software.	Drive is faulted.	By user: switch to inspection/RDF mode or cycle power.	Disconnect main contactor status signal.
	2016	0	0	Main contactor stuck	Main contactor status indicates that contactor has not been opened between runs.	Main contactor or auxiliary contact broken. Problem with status wiring.	Check wiring. Check main contactor. Check auxiliary contactor block.	Drive monitors main contactor status input by software.	Drive is faulted.	Automatically when main contactor status input works.	Short main contactor status signal during run.
	2017	0	0	Line bridge fault	Line bridge does not start.	Line bridge does not start.	Check other fault codes. Check line voltages and network stability during run. Check parameters: - start delay (33) Check K201, K201:3, K201:4 and wiring. Cycle power.	Drive tries to start 5 times and if line bridge does not start supervision is tripped.	Drive is faulted.	By user: switch to inspection/RDF mode or cycle power.	-
	2020	0	0	Modulation disabled	Line or motor bridge has stopped modulation without permission.	201 contactor status input drops during run.	Check 201 contactor and aux contact block. Check 201 aux contact block wiring. If problem persist update drive sw.	Drive stops if PWM ENABLE or KCDI status signals, STATUS_ON or STATUS_RUN, change state without permission during run.	Stops immediately.	Automatic. If wiring checked, update software. In case problem persists, replace drive.	Disconnect PWM ENABLE from drive during run.
	2079	0	0	Brake controller fault	Brake controller safety chain feedback signal to drive CPU stuck to closed state at start.	Broken drive control board. Broken brake control board. Broken cable between DCBG and brake control board.	Check brake controller power supplies. Check cables from DCBG to brake controllers.	Drive monitors "drive enable" signal from brake controller, the signal indicates the status of safety chain input to brake controller. If the signal status is not correct then fault is set and run aborted.	Drive is faulted.	Automatically when safety chain input works.	Connect pin 6 on 10-pin cable between DCBG and BCX08 to ground.
	2107	0	0	KDM Drive invalid status fault	Invalid KCDI status detedted while standing.	KDM Drive serial link XRS1 status wrong.	Check and update KDM Drive software.	KCDI status bits (STATUS_ON or STATUS_RDY) are high for longer than 2 seconds while standing.	Drive is faulted.	Automatic.	Check KDM Drive software.
121 Command supervision											
	3085	0	0	PES (Performance selection) parameter check							
122 Drive time supervision											
	1004	0	0	Drive Time Supervision	Drive time exceeded.	Drive time exceeded calculated maximum time set for the drive.	Turn elevator to RDF or cycle power. Check car cable signals and connectors. Check mecanical installation. Check elevator balancing.	Drive software monitors drive time.	Drive faulted after exceeded 3 times in row.	Automatic.	
123 Motor parameters											
	1007	0	0	Motor parameterization fault	Motor parameters not valid.	One or more of the following parameters is not set: - Motor setup type (6_60) - Motor source voltage (6_80) - Motor nominal current (6_81) - Motor nominal stator frequency (6_82) - Motor nominal speed (6_83) - Motor nominal output power (6_84)	Set motor parameters. - Motor setup type (6_60) - Motor source voltage (6_80) - Motor nominal current (6_81) - Motor nominal stator frequency (6_82) - Motor nominal speed (6_83) - Motor nominal output power (6_84)	Drive software monitors parameters.	Drive is faulted.	By user: enter valid motor parameters.	Set a motor parameter to zero and switch elevator to normal mode.
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124 Sensors and wiring											
	1012	0	0	77 switch fault	77:U/N active at the same time or 77:U/N active at wrong place.	77:U/N active at the same time or 77:U/N active at wrong place. Broken/malfunctioning 77 switch or problem with switch cabling.	Check LCECPU <=> LCEDCBG cable and connectors. Check LCECPU <=> car top cross connection box cables and connectors. Check 77:U/N switches and cables and connectors. Check switch reader fixing.	Drive monitors 77 switch information and if both 77:U and 77:N are active at the same time or 77:U/N is active at wrong place supervision is tripped.	Drive is faulted.	Automatically when switch information is correct or by user: switch to RDF/inspection mode.	Set both 77 switches on at the same time.
	1013	0	0	77S switch fault	77S stuck	77S stuck. Broken/malfunctioning 77S switch or problem with switch cabling.	Check LCECPU <=> LCEDCBG cable and connectors. Check LCECPU <=> car top cross connection box cables and connectors. Check 77:U/N/S switches and cables and connectors. Check switch reader fixing.	Drive monitors 77S and if it is on elsewhere than terminal floors supervision is tripped.	Drive is faulted at floor.	Automatically when switch information is correct or by user: switch to RDF/inspection mode.	Set 77S on continuously.
	1032	0	0	77S switch fault	77S missing	77S missing. Broken/malfunctioning 77S switch or problem with switch cabling.	Check LCECPU <=> LCEDCBG cable and connectors. Check LCECPU <=> car top cross connection box cables and connectors. Check 77:U/N/S switches and cables and connectors. Check switch reader fixing.	Drive monitors 77S and if it is not on in terminal floors supervision is tripped.	Drive is faulted at floor.	Automatically when switch information is correct or by user: switch to RDF/inspection mode.	Disable 77S switch.
	1054	0	0	SD card supervision	Normal mode is not allowed if SD card is in the reader.	SD card can only be used for sw update and must be removed after the update is ready.	Remove SD card from LCEDCBG board.	Drive supervises SD card socket.	Only inspection run is possible.	Automatic, after SD card is removed from reader.	Insert SD card and set elevator to normal mode.
	2057	0	0	Motor encoder fault	Speed detected from motor without run command or motor encoder is faulted.	Motor encoder not connected. Motor encoder broken. Problem with motor encoder wiring.	Check motor encoder wiring and connector(s). Replace motor encoder.	Drive supervises motor encoder pulses and calculated speed.	Stops immediately and drive is faulted.	Automatic	Disconnect encoder.
	2105	0	0	Brake status failed (moving)	Drive detected that brake status signal is in wrong state while running.	Brake controller failed or broken wire.	Check wiring from DCBG to brake controller. Check brake controller.	Drive supervises brake status signal during the run and if its status is not correct supervision is tripped	Stops immediately.	Automatic.	Disconnect XB1 pin 4 while running.
	3060	position [cm]	speed [cm/s]	NTS switch position	Detected NTS switch position does not match with setup.	Detected NTS switch (77Ux or 77Nx) position does not match with setup. Problem with motor encoder, bad NTS setup or problem with NTS switches.	Check shaft switch 77:Ux and 77:Nx positions. Check 77:Ux and 77:Nx switches. Check magnet switch and reader installation according to shaft vane diagram. Avoid magnet installation close to guide rail fixing bolts. Make new shaft setup.	NTS function compares detected switch positions between saved positions and in case of too large error supervision is tripped. NTS-led is blinks on DCBG board.	Ramp stop to next floor.	Automatic	Move NTS switch(es).
	3061	position [cm]	speed [cm/s]	NTS switch order	Wrong NTS switch order.	Wrong NTS (77Ux or 77Nx) switch order. Problem with NTS switches: 77Ux and 77Nx	Check shaft switch 77:Ux and 77:Nx positions. Check 77:Ux and 77:Nx switches. Check magnet switch and reader installation according to shaft vane diagram. Avoid magnet installation close to guide rail fixing bolts. Make new shaft setup.	NTS function supervises that switches appear in correct sequence. In case of error supervision is tripped. NTS-led is blinks on DCBG board.	Ramp stop to next floor.	Automatic	Swap NTS switches.

	3064	2=missing files 3=checksum 4=SD card type 5=old SD card	0	SD card warning	Drive cannot read SD card.	Drive cannot read SD card. SD card is corrupted, empty or does not have correct files.	Check SD card files. Use only KONE official SD card from Global Spares Supply.	Drive identifies SD card when it is inserted to reader.	SD card cannot be used for sw update.	Automatic	Insert empty SD card.
	3068	position [cm]	speed+mode [cm/s] + last number is mode	Drive - LCE communication fault	Communication link between LCE CPU and DCBG has failed for a moment.	Broken wiring, disturbance, grounding error, control system failure. Broken connector pin.	Check wiring, grounding and shielding between LCECPU and Drive. Check grounding and shielding between drive and machinery.	Communication link is checked continuously, timeout is 500ms. NTS-led is lit on DCBG board.	Ramp stop to next floor if main contactor stays on.	Automatic	Cause disturbance to the communication link
Main code	**Sub code**	**Data 1**	**Data 2**	**Name**	**Description**	**Reason**	**Action**	**Detection**	**Operation**	**Recovery**	**Testing**
126 Motor brakes and brake controller											
[redacted]	1040	1 = brake switch input 1 is in picked state 2 = brake switch input 2 is in picked state 3 = both input are in picked state	0	Brake feedback fault (pick)		1 = brake switch input 1 is in picked state 2 = brake switch input 2 is in picked state 3 = both input are in picked state	Check brake switch cables. Check brake switches. Check brakes. Check brake wiring.				
	1045 A	0	0	Brake current with reduced voltage supervision counter	Detected brake current with reduced voltage failed too many times.	Too many 3059 warnings.	Check that brake coils have separated wiring. Check brake pick and hold voltage parameters (6_90, 6_91).	This fault occurs if there's too many consecutive 126_3059 warnings.	Drive is faulted.	By user: switch to inspection/RDF mode.	Repeat testing part of case 126_3059 3 times.
	1046	Measured drift in mm	0	Automatic brake test	Automatic brake test has failed 3 times in a row.	Too many 2072 faults.	Check brake wiring and operation with manual brake test. Check brakes according to AS instruction.	This fault occurs if there's too many consecutive 126_2071 or 126_2072 faults.	Drive is faulted.	By user: switch to inspection/RDF mode.	Perform manual brake test 3 times with movement.
	1047	0	0	Too low brake current supervision counter	No brake current too many times.	Too many 2104 faults.	Check brake wiring and connectors. Measure that brake coil resistance is not infinite. Check brake pick voltage parameter (parameter 6_90). Check brake current settings S1 and S2 in KDHBCM.	This fault occurs if there's too many consecutive 126_2104 faults.	Drive is faulted.	By user: switch to inspection/RDF mode.	Disconnect both brake cables from brake control module and try to drive.
	1058	0	0	BCX brake controller communication fault	No communication from BCX to DCBG for too long.	Communication from BCX to DCBG is disturbed during run.	Check brake cable shielding. Check drive and motor groundings.	This fault occurs when BCX feedback is missing for too long.	Drive is faulted	By user: switch to inspection/RDF mode.	Drive the elevator, wait at least 3 seconds and disconnect current feedback from XB1. Elevator should run at least 21 seconds for the fault to occur.
	1059	0	0	BCX brake controller communication fault	No correct reference messages from DCBG to BCX for too long.	Communication from DCBG to BCX is disturbed during run.	Check brake cable shielding. Check drive and motor groundings.	This fault occurs when BCX is not receiven a correct message for too long.	Drive is faulted	By user: switch to inspection/RDF mode.	Drive the elevator, wait at least 3 seconds, create disturbance on voltage reduction signal in XB1 so that message cannot be correctly received from BCX. Elevator should run at least 21seconds for the fault to occur.
	1060	BCX fault codes in field Data 1: 1 = +24V supply is below 10V 2 = Relay(s) are stuck 4 = Short circuit in output 8 = Brake control signal stuck high 16 = Coil leaks to ground 32 = Too low brake current 64 = CRC16/RAM Test/Register failed 128 = Brake signal possibly stuck low 256 = Drive enable signal is stuck (code is sum of previous in case of multiple faults. Example 6 = Relays are stuck AND Short circuit in output)		Brake control board (BCX) specific failure feedback	Brake controller has detected fault fault 3 times in row..	Too many 3075 faults.	Check brake wiring and connectors. Measure that brake coil resistance is not: - infinite - close to 0 - shorted to ground Check brake pick voltage parameter (parameter 6_90) Check drive XS3 connector and wiring.	This fault occurs if there's too many consecutive 126_3075 faults.	Drive is faulted.	By user: switch to inspection/RDF mode.	Arrange one BCX specific harware failure
		600	0	CPU overloaded / busy	Some task is taking too long to execute.	CPU doesn't have enough time to execute all tasks within set time period.	If problem persists then replace the BCX14 brake control module.	Background task loop counter is not increasing between higher priority interrupt cycle(s).	Brake operation disabled.	Power on-off-on required. If problem remains, replace brake module.	
		601	item	Software error detected (RAM corrupted).	Variable in CPU memory has detected to have corrupted value.	Variable in CPU memory has detected to have corrupted value: 0 = rom check state 1 = application state 2 = ram check state 3 = brake control state 4 = communication read state 5 = communication write state	If problem persists then replace the BCX14 brake control module.	Most critical variables are being range checked and some are checked against secondary instance of variable to have valid value.	Brake operation disabled.	Power on-off-on required. If problem remains, replace brake module.	
		602	0	Rom checksum failure	Pre-calculated checksum does not match with one calculated.	Pre-calculated checksum does not match with one calculated.	If problem persists then replace the BCX14 brake control module.	Checksum is being calculated at power up and during run time at minimum once per day. It must match with pre-calculated checksum stored onto the flash.	Brake operation disabled.	Replace brake module.	
		603	0/1	Ram checksum failure	Ram memory area read/write verify failed.	Ram memory area read/write verify failed.	If problem persists then replace the BCX14 brake control module.	Whole Ram memory area is being tested at power up and at minimum interval once per day.	Brake operation disabled.	Power on-off-on required. If problem remains, replace brake module.	
		604	8	Startup sequence fault	Brake open startup sequence was not finished.	Brake startup sequence was not completed within time.	If problem persists then replace the BCX14 brake control module.	1. Ac present must be detected 2. To enable PFC, UDC voltage must be higher than 240V 3. To enable operation, UDC voltage must be higher than 300V while PFC is enabled	Brake will not open.	Measure 230V supply voltage from XB11:1 - XB11:2 when main contactor is energized. If voltage ok, replace brake module. If not, check fuse F?.	
		605	0.01V	24VDc supply fault	Supply voltage below 16V.	Supply voltage below 16V.	Check control power supply cable from DCBG to BCX14.	Measured supply voltage is measured to be below 16V.	Brake operation disabled.	Automatically when supply voltage is above 17V. Power on-off-on to check if fault was caused by multifuse. Measure 24V supply from KDM XOPT1:1 - XOPT1:5 and BCX14 XOPT1:1 - XOPT1:2. If above 17V, replace brake module. If not ok, check KDM board supply.	
		606	100mA	Brake current follow fault	Measured current doesn't match current order.	Measured brake current does not follow reference.	Check brake wiring and operation with manual brake test. Check brake pick current (6_90) and brake hold current (6_91).	Measured current doesn't match current order within 500mA maximum difference.	Brake operation disabled.	Automatically. Check brake parameters (6_90, 6_91). Check brake connections and measure brake resistance (compare). If problem continues, replace module.	
		607	0	Electronic name plate checksum failure	No electronic name plate set or electronic name plate stored is corrupted.	Electronic name plate must be set for new boards. In case electronic name plate exists, but is corrupted, board must be replaced.	If problem persists then replace the BCX14 brake control module.	Name plate is read at power up and calculated checksum is compared to one stored with the name plate.	Brake operation disabled.	Replace brake module.	
		608	0	Major software version change detected	Electronic name plate major version doesn't match with active software.	Software version changed unintentionally.	If problem persists then replace the BCX14 brake control module.	Name plate is read at power up and version information is compared to active software. Minor software version update is allowed without updating name plate.	Brake operation disabled.	Replace brake module.	
		609	0	IGBT fault	IGBT transistor current too high	Short circuit on brake output.	Check brake wiring and operation with manual brake test.	Voltage across the igbt device is measured and if it is too high, IGBT fault signal is activated	Brake operation disabled.	Measure brake resistance (compare). Check brake cable (cable length, connections, insulation resistance). If values are ok, replace brake module.	

Main code	Sub code	Data 1	Data 2	Name	Description	Reason	Action	Detection	Operation	Recovery	Testing
		610	0/1	Over current measured	Measured current is above 5A.	Measured current is above 5A. 0 = Brake A 1 = Brake B	Check brake wiring and operation with manual brake test. If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Check brake parameters (6_90, 6_91). Check brake connections and measure brake resistance (compare). If problem continues, replace module.	
		611	item	Internal BCX14 brake transistor status fault	Invalid board internal brake switch status at brake open. Short circuit on brake output.	Invalid brake transistor status at brake open: 0 = Brake A, phase 1 1 = Brake B, phase 1 128 = Brake A, phase 2 129 = Brake B, phase 2	Check brake wiring and operation with manual brake test. If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Check brake cable (connections, insulation resistance). Measure brake resistance (compare). If values are ok, replace brake module.	
		612	item	Earth leagake detected	Current measured too early on opening brakes.	Current detected at brake open: 0 = Brake A, phase 1 1 = Brake B, phase 1 128 = Brake A, phase 2 129 = Brake B, phase 2	Check brake wiring and operation with manual brake test. If problem persists then replace the BCX14 brake control module.	Measured current is above 300mA, while brake output control 0/1 = low side is kept high (phase 1) 128/129 =high side is kept high (phase 2)	Brake operation disabled.	Check brake cable (connections, insulation resistance). Measure brake resistance (compare). If values are ok, replace brake module.	
		613	c	CPU over temperature	CPU temperature too high.	Measured CPU temperature exceeds it's set maximum limit (120 c).	If problem persists then replace the BCX14 brake control module.	Temperature is read using CPU internal temperature sensor.	Brake operation disabled.	Automatically. Check brake module environment temperature. If probelm continues, replace brake module.	
		614	cnt	VRED communication fault	VRED communication (reference message) is corrupted.	Message timing doesn't match causing invalid reference message to be read. Previous reference is used.	Check brake cable shielding. Check drive and motor groundings.	Message should match it's predefined pattern. Fault is risen on 10 invalid messages received in row.	Brake operation disabled.	Automatically. Check XB4 wiring and connection (KDM-BCX14). If problem continues, replace brake module or KDM board.	
		615	0	Ram stack low detected	Free Ram stack area is too low.		If problem persists then replace the BCX14 brake control module.	Ram free stack area is being monitored while running. If free stack area goes below it's limits, fault is risen.	Brake operation disabled.	Power on-off-on required. If problem remains, replace brake module.	
		616	item	UDC voltage fault	Measured UDC voltage has exceeded it's set limits.	UDC voltage fault: 2 = Below 300V while PFC is enabled 4 = Below 240V while PFC disabled 5 = Over 450V	Check BCX14 brake controller supply connector XB11. If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Automatically. Measure 230V supply voltage from XB11:1 - XB11:2 when main contactor is energized. If values ok, replace brake module.	
		617	0	Eeprom communication fault	Eeprom write operation has failed.	Eeprom chip did not send valid response to write operation.	If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Power on-off-on required. If problem remains, replace brake module.	
		618	0	Brake not connected	Brake not connected.	Brake not connected: 0 = Brake A 1 = Brake B	Connect brakes. If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Check brake cable (connections, insulation resistance). Measure brake resistance (compare). If values are ok, replace brake module.	
		619	0	Brake enable before AC present.			Check K201, K201:3 and K201:4 wiring. If problem persists then replace the BCX14 brake controller.		Brake operation disabled.	Automatically. Measure 230V supply voltage from XB11:1 - XB11:2 when main contactor is energized. If values ok, replace brake module.	
		620	0	Charge fault	Over current of charge fet.	Charge fet is turned on before DC-link has charged.	Check K201, K201:3 and K201:4 wiring. If problem persists then replace the BCX14 brake controller.	Voltage across the mosfet transistor is measured and if it is too high, charge fault signal is activated	Brake operation disabled.	Automatically. Measure 230V supply voltage from XB11:1 - XB11:2 when main contactor is energized and brakes are opening. If values ok, replace brake module.	
		621	0	Invalid PA status		PA status not high while AC present is detected, or PA status not low while no AC present.	If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Automatically when status is correct. If problem persists, replace brake module.	
	1063	0	0	Brake status failed	Brake status signal failed too many times	Problem with brakes, broken wire in connector XB1 or wrong system parameterization for parameter 6_61.	Check cables from DCBG to brake controllers.	Drive supervises brake status signal during the run and if its status is not correct for three consecutive times supervision is tripped	Drive is faulted.	By user: switch to inspection/RDF mode.	Disconnect XB1 pin 4.
	2071 A	Measured drift in mm	Speed in mm/sec when test ends	Brake test failed: speed	Speed detected from motor.	Movement detected from motor encoder. Problem with brakes or brake wiring or elevator balancing (one brake cannot hold elevator in place).	Check brake wiring and operation with manual brake test. Check brakes according to AS instruction. Check elevator balancing. For NanoSpace only: check parameter 6_36 (brake test torque)	Drive supervises speed during automatic brake test. This supervision is active only if automatic brake test is selected in use.	Drive is faulted.	Automatic	During brake test rotate motor or motor speed sensor.
	2072 A	Measured drift in mm	Speed in mm/sec when test ends	Brake test failed: position	Movement detected from car.	Movement detected from motor encoder. Problem with brakes or brake wiring or elevator balancing (one brake cannot hold elevator in place).	Check brakes according to AS instruction. Check brake wiring and operation with manual brake test. Check elevator balancing. For NanoSpace only: check parameter 6_36 (brake test torque)	Drive supervises elevator position (61:U/N) during automatic brake test. This supervision is active only if automatic brake test is selected in use.	Drive is faulted.	Automatic	During brake test move elevator car so that either signal 61:U/N disappears.
	2104	0	0	Too low brake current	No brake current detected when opening brakes.	Too low brake current detected when opening brakes. Problem with brakes, wiring or with brake control board.	Check brake wiring and connectors. Measure that brake coil resistance is not infinite. Check brake pick voltage parameter (parameter 6_90) Check brake current settings S1 and S2 in KDHBCM. Check drive XS3 connector and wiring.	Drive monitors brake current and if current is not detected supervision is tripped.	Drive is faulted.	Automatic	Disconnect brake cable from brake control module and try to drive.
	2108	0	0	Single brake stopping from nominal speed: brake 1 test failed	Brake 1 was not producing adequate torque to decelerate the car from a nominal speed quick stop	Deceleration from full speed with single brake only is not according EN81-20	Check brake wiring and operation with manual brake test. Check brakes according to AS instruction. Check elevator balancing.	Drive monitors the deceleration rate during a quickstop test performed with a single brake from nominal speed in the middle of the shaft. If the deceleration does not satisfy a predefined limit, test is failed and fault code is issued.	Drive is faulted.	Automatic	
	2109	0	0	Single brake stopping from nominal speed: brake 2 test failed	Brake 2 was not producing adequate torque to decelerate the car from a nominal speed quick stop	Deceleration from full speed with single brake only is not according EN81-20	Check brake wiring and operation with manual brake test. Check brakes according to AS instruction. Check elevator balancing.	Drive monitors the deceleration rate during a quickstop test performed with a single brake from nominal speed in the middle of the shaft. If the deceleration does not satisfy a predefined limit, test is failed and fault code is issued.	Drive is faulted.	Automatic	
	3058 A	0	0	Too low brake current warning	No brake current detected when opening brake or BCX pick fault	Too low brake current detected when opening brakes. Problem with brakes, wiring or with brake control board.	Check brake wiring and connectors. Measure that brake coil resistance is not infinite. Check brake pick voltage parameter (parameter 6_90) Check drive XS3 connector and wiring.	Drive monitors brake current and if current is not detected or BCX reports a pick fault, supervision is tripped.	Drive is faulted.	Automatic	Disconnect brake cable from brake control module and try to drive, or set low pick voltage and try to drive.
	3059 A	0	0	Brake current with reduced voltage warning	Detected brake current with reduced voltage is too high.	Problem with brake cables or with brake controller. Problem with brake control parameters.	Check brake cables. Check brake control parameters: - Brake control type (parameter 6_61) - Brake pick voltage (parameter 6_90) - Brake hold voltage (parameter 6_91)	Drive monitors reduced voltage brake current and if current will not be smaller than with full voltage supervision is tripped. Supervision is active only when reduced voltage is selected in use.	Drive is faulted.	Automatic	Set parameter 6_61 to 1 and feed AC current through brake controller board's current sensor when elevator runs.

	3075	BCX fault code: 1 = +24V supply is below 10V 2 = Relay(s) are stuck 4 = Short circuit in output 8 = Brake control signal stuck high 16 = Coil leaks to ground 32 = Too low brake current 64 = CRC16/RAM Test/Register failed 128 = Brake signal possibly stuck low 256 = Drive enable signal is stuck (code is sum of previous in case of multiple faults. Example 6 = Relays are stuck AND Short circuit in output)		Brake control board (BCX) specific failure feedback	Brake controller has detected fault.	BCX reports an hardware failure. Problem with brakes or with brake control board.	Check brake wiring and connectors. Measure that brake coil resistance is not: - infinite - close to 0 - shorted to ground Check brake pick voltage parameter (parameter 6_90) Check drive XS3 connector and wiring.	Drive monitors faults detected by the brake controller. Brake controller faults are reported via the serial communication link (VRED).	Drive is faulted.	Automatic	Set up a fault for BCX.
Main code	**Sub code**	**Data 1**	**Data 2**	**Name**	**Description**	**Reason**	**Action**	**Detection**	**Operation**	**Recovery**	**Testing**
		600	0	CPU overloaded / busy	Some task is taking too long to execute.	CPU doesn't have enough time to execute all tasks within set time period.	If problem persists then replace the BCX14 brake control module.	Background task loop counter is not increasing between higher priority interrupt cycle(s).	Brake operation disabled.	Power on-off-on required. If problem remains, replace brake module.	
		601	item	Soft error detected (RAM corrupted).	Variable in CPU memory has detected to have corrupted value.	Variable in CPU memory has detected to have corrupted value: 0 = rom check state 1 = application state 2 = ram check state 3 = brake control state 4 = communication read state 5 = communication write state	If problem persists then replace the BCX14 brake control module.	Most critical variables are being range checked and some are checked against secondary instance of variable to have valid value.	Brake operation disabled.	Power on-off-on required. If problem remains, replace brake module.	
		602	0	Rom checksum failure	Pre-calculated checksum does not match with one calculated.	Pre-calculated checksum does not match with one calculated.	If problem persists then replace the BCX14 brake control module.	Checksum is being calculated at power up and during run time at minimum once per day. It must match with pre-calculated checksum stored onto the flash.	Brake operation disabled.	Replace brake module.	
		603	0/1	Ram checksum failure	Ram memory area read/write verify failed.	Ram memory area read/write verify failed.	If problem persists then replace the BCX14 brake control module.	Whole Ram memory area is being tested at power up and at minimum interval once per day.	Brake operation disabled.	Power on-off-on required. If problem remains, replace brake module.	
		604	8	Startup sequence fault	Brake open startup sequence was not finished.	Brake startup sequence was not completed within time.	If problem persists then replace the BCX14 brake control module.	1. Ac present must be detected 2. To enable PFC, UDC voltage must be higher than 240V 3. To enable operation, UDC voltage must be higher than 300V while PFC is enabled	Brake will not open.	Measure 230V supply voltage from XB11:1 - XB11:2 when main contactor is energized. If voltage ok, replace brake module. If not, check fuse F?.	
		605	0.01V	24VDc supply fault	Supply voltage below 16V.	Supply voltage below 16V.	Check control power supply cable from DCBG to BCX14.	Measured supply voltage is measured to be below 16V.	Brake operation disabled.	Automatically when supply voltage is above 17V. Power on-off-on to check if fault was caused by multifuse. Measure 24V supply from KDM XOPT1:1 - XOPT1:5 and BCX14 XOPT1:1 - XOPT1:2. If above 17V, replace brake module. If not ok, check KDM board supply.	
		606	100mA	Brake current follow fault	Measured current doesn't match current order.		Check brake wiring and operation with manual brake test. Check brake pick current (6_90) and brake hold current (6_91).	Measured current doesn't match current order within 500mA maximum difference.	Brake operation disabled.	Automatically. Check brake parameters (6_90, 6_91). Check brake connections and measure brake resistance (compare). If problem continues, replace module.	
		607	0	Electronic name plate checksum failure	No electronic name plate set or electronic name plate stored is corrupted.	Electronic name plate must be set for new boards. In case electronic name plate exists, but is corrupted, board must be replaced.	If problem persists then replace the BCX14 brake control module.	Name plate is read at power up and calculated checksum is compared to one stored with the name plate.	Brake operation disabled.	Replace brake module.	
		608	0	Major software version change detected	Electronic name plate major version doesn't match with active software.		If problem persists then replace the BCX14 brake control module.	Name plate is read at power up and version information is compared to active software. Minor software version update is allowed without updating name plate.	Brake operation disabled.	Replace brake module.	
		609	0	IGBT fault	IGBT transistor current too high	Short circuit on brake output.	Check brake wiring and operation with manual brake test.	Voltage across the igbt device is measured and if it is too high, IGBT fault signal is activated	Brake operation disabled.	Measure brake resistance (compare). Check brake cable (cable length, connections, insulation resistance). If values are ok, replace brake module.	
		610	0/1	Over current measured	Measured current is above 5A.	Measured current is above 5A. 0 = Brake A 1 = Brake B	Check brake wiring and operation with manual brake test. If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Check brake parameters (6_90, 6_91). Check brake connections and measure brake resistance (compare). If problem continues, replace module.	
		611	item	Brake switch status fault	Invalid board internal brake switch status at brake open. Short circuit on brake output.	Invalid brake switch status at brake open: 0 = Brake A, phase 1 1 = Brake B, phase 1 128 = Brake A, phase 2 129 = Brake B, phase 2	Check brake wiring and operation with manual brake test. If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Check brake cable (connections, insulation resistance). Measure brake resistance (compare). If values are ok, replace brake module.	
		612	item	Earth leagake detected	Current measured too early on opening brakes.	Current detected at brake open: 0 = Brake A, phase 1 1 = Brake B, phase 1 128 = Brake A, phase 2 129 = Brake B, phase 2	Check brake wiring and operation with manual brake test. If problem persists then replace the BCX14 brake control module.	Measured current is above 300mA, while brake output control 0/1 = low side is kept high (phase 1) 128/129 =high side is kept high (phase 2)	Brake operation disabled.	Check brake cable (connections, insulation resistance). Measure brake resistance (compare). If values are ok, replace brake module.	
		613	c	CPU over temperature	CPU temperature too high.	Measured CPU temperature exceeds it's set maximum limit (120 c).	If problem persists then replace the BCX14 brake control module.	Temperature is read using CPU internal temperature sensor.	Brake operation disabled.	Automatically. Check brake module environment temperature. If probelm continues, replace brake module.	
		614	cnt	VRED communication fault	VRED communication (reference message) is corrupted.	Message timing doesn't match causing invalid reference message to be read. Previous reference is used.	Check brake cable shielding. Check drive and motor groundings.	Message should match it's predefined pattern. Fault is risen on 10 invalid messages received in row.	Brake operation disabled.	Automatically. Check XB4 wiring and connection (KDM-BCX14). If problem continues, replace brake module or KDM board.	
		615	0	Ram stack low detected	Free Ram stack area is too low.		If problem persists then replace the BCX14 brake control module.	Ram free stack area is being monitored while running. If free stack area goes below it's limits, fault is risen.	Brake operation disabled.	Power on-off-on required. If problem remains, replace brake module.	
		616	item	UDC voltage fault	Measured UDC voltage has exceeded it's set limits.	UDC voltage fault: 2 = Below 300V while PFC is enabled 3 = Below 240V while PFC disabled 4 = Over 450V	Check BCX14 brake controller supply connector XB11. If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Automatically. Measure 230V supply voltage from XB11:1 - XB11:2 when main contactor is energized. If values ok, replace brake module.	
		617	0	Eeprom communication fault	Eeprom write operation has failed.	Eeprom chip did not send valid response to write operation.	If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Power on-off-on required. If problem remains, replace brake module.	

Main code	Sub code	Data 1	Data 2	Name	Description	Reason	Action	Detection	Operation	Recovery	Testing
		618	0	Brake not connected	Brake not connected.	Brake not connected: 0 = Brake A 1 = Brake B	Connect brakes. If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Check brake cable (connections, insulation resistance). Measure brake resistance (compare). If values are ok, replace brake module.	
		619	0	Brake enable before AC present.			Check K201, K201:3 and K201:4 wiring. If problem persists then replace the BCX14 brake controller.		Brake operation disabled.	Automatically. Measure 230V supply voltage from XB11:1 - XB11:2 when main contactor is energized. If values ok, replace brake module.	
Main code	**Sub code**	**Data 1**	**Data 2**	**Name**	**Description**	**Reason**	**Action**	**Detection**	**Operation**	**Recovery**	**Testing**
		620	0	Charge fault	Over current of charge fet.	Charge fet is turned on before DC-link has charged.	Check K201, K201:3 and K201:4 wiring. If problem persists then replace the BCX14 brake controller.	Voltage across the mosfet transistor is measured and if it is too high, charge fault signal is activated	Brake operation disabled.	Automatically. Measure 230V supply voltage from XB11:1 - XB11:2 when main contactor is energized and brakes are opening. If values ok, replace brake module.	
		621	0	Invalid PA status		PA status not high while AC present is detected, or PA status not low while no AC present.	If problem persists then replace the BCX14 brake control module.		Brake operation disabled.	Automatically when status is correct. If problem persists, replace brake module.	
	3083	1 = brake switch input 1 is in drop state 2 = brake switch input 2 is in drop state 3 = both inputs are in drop state	0	Brake switch fault (dropping)		Drive CPU monitors brake switch inputs during run. Supervision is tripped if one of the brake switch inputs remains in drop state ~2 seconds after brake pick command.	Check brake switch cables. Check brake switches. Check brakes. Check brake wiring. Check brake controller.				
	6021	Measured drift in mm	Speed in mm/sec when test ends	Brake 1 test passed info	Reports success of brake 1 manual test.	1 = brake switch input 1 is in drop state	-	Drive determines whether test was successfull or not.	Code is shown after successful test.	-	Perform manual brake test.
	6022	Measured drift in mm	Speed in mm/sec when test ends	Brake 2 test passed info	Reports success of brake 2 manual test.	2 = brake switch input 2 is in drop state	-	Drive determines whether test was successfull or not.	Code is shown after successful test.	-	Perform manual brake test.
	6023 (A)	Measured drift in mm	Speed in mm/sec when test ends	Automatic brake test is passed info	Reports success of automatic brake test	3 = both inputs are in drop state	-	Drive determines whether test was successfull or not.	Code is shown after successful test.		Perform automatic brake test with value (3).
	6024	0	0	Single brake stopping from nominal speed: brake 1 test passed info	Reports that Brake 1 successfully decelerated the car from a nominal speed quick stop test	User has performed single brake 1 stopping test (6_72_25) from nominal speed.	-	Drive monitors the deceleration rate during a quickstop test performed with a single brake from nominal speed in the middle of the shaft. If the deceleration satisfies a predefined limit, test is passed and info code is issued.	Drive is faulted.	Automatic	Perform single brake stopping from nominal speed test 6_72_25.
	6025	0	0	Single brake stopping from nominal speed: brake 2 test passed info	Reports that Brake 2 successfully decelerated the car from a nominal speed quick stop test	User has performed single brake 2 stopping test (6_72_26) from nominal speed.	-	Drive monitors the deceleration rate during a quickstop test performed with a single brake from nominal speed in the middle of the shaft. If the deceleration satisfies a predefined limit, test is passed and info code is issued.	Drive is faulted.	Automatic	Perform single brake stopping from nominal speed test 6_72_25.
Main code	**Sub code**	**Data 1**	**Data 2**	**Name**	**Description**	**Reason**	**Action**	**Detection**	**Operation**	**Recovery**	**Testing**
130 Drive codes											
	5000	1	0	Motor bridge overcurrent	Measured motor current is too high.	Brakes do not open, mechanical problem hindering movement, wrong motor parameters in group 6_80-6_85, broken motor encoder, short-circuit or earth fault in motor cables or broken motor current sensor.	Check elevator parameters (6_1, ..., 6_14). Check machinery parameters (6_60, ..., 6_65 and 6_80, ..., 6_91) Check brakes. Check shaft mechanics. Check motor wiring. Check motor connectors. Check drive connectors.	Motor current is supervised by a hardware sensors. Supervision trips if overcurrent limit is exceeded.	Stops immediately.	Automatically when fault disappears.	Short-circuit two output phases together, and try to run the elevator.
		2	0	DC-link overvoltage	Measured DC-link voltage is too high.	Braking chopper is used but braking resistors are not connected, high overvoltage spikes detected in the supply network, line bridge does not function properly, supply network has disappeared when elevator has moved to light direction.	Check DC link voltage from drive real time display (6_75_110). Check power supply lines. Check elevator features (BMV, EBD, EPD). Check elevator fuses L1, L2, and L3. Check other fault codes.	Voltage over DC-link capacitors is measured. Supervision trips if the voltage is higher than 910 V.	Stops immediately.	Automatically when fault disappears.	Disable line bridge control, and try to run the elevator to light direction.
		7	0	Motor bridge saturation	Voltage generated by the motor bridge does not follow the reference.	Motor bridge is broken.	If problem persistst change drive (KR6 or KR7) or INU (KR8).	Drive internal supervision.	Stops immediately.	Replace drive.	
		9	0	DC-link undervoltage	Measured DC-link voltage is too low.	Main fuse is blown, one input phase is missing, supply network voltage is too low, problem in the charging circuit or the drive is broken.	Check DC link voltage from drive real time display (6_75_110). Check power supply lines. Check elevator features (BMV, EBD, EPD). Check elevator fuses L1, L2, and L3. Check other fault codes.	Voltage over DC-link capacitors is measured. Supervision trips if the measured voltage is lower than 263 V.	Stops immediately and drive is faulted.	Automatically when fault disappears.	Disconnect one of the supply phases.
		10	0	Input phase unbalance	Too big voltage unbalance between input phases.	Main contactor poles opened unevenly, supply fuse blown, supply wiring loose, too big unbalance between supply voltages.	Check DC link voltage from drive real time display (6_75_110). Check power supply lines. Check elevator features (BMV, EBD, EPD). Check elevator fuses L1, L2, and L3. Check other fault codes.	Drive internal software supervision.	Stops immediately.	Automatically when fault disappears.	Disconnect one of the supply phases when elevator is running.
		11	0	Output phase missing	Current measurement has detected that there is no current in one motor phase.	One or more motor wires are not connected to the drive, or there is a wiring problem in the motor itself.	Check motor supply wires. Check motor supply plugs.	Motor currents are measured. Supervision trips if the sum does not equal zero within the limits.	Stops immediately.	Automatically when fault disappears.	Disconnect one of the motor phases and try to run the elevator.
		13	0	Motor bridge undertemp	Heatsink temperature is too low.	Drive temperature is below operational limits.	Check machine room temperature.	Heatsink temperature is measured. Supervision trips if the temperature is below -10 C.	Stops immediately.	Automatically when heatsink temperature is high enough.	Cool heatsink.
		14	0	Motor bridge overtemp	Heatsink temperature is too high.	Drive temperature is above operational limits, excessive load on drive, drive fan is not operating, ambient temperature is too high, switching frequency of motor bridge's transistors is too high.	Check drive fan (KR6 or KR7) or INU fan (KR8) operation. Check drive heatsink (KR6 or KR7) or INU heatsink (KR8) airflow. Check shaft or machine room temperature. Check RT display for drive heatsink temperature (6_75_40).	Heatsink temperature is measured. Supervision trips if the temperature is above 100 C.	Stops immediately.	Automatically when heatsink temperature is low enough.	Heat heatsink.
		25	0	Start angle not found	PMSM: direction of rotor's north pole was not found.	Motor brakes are not holding during angle detection, wrong motor parameters in group 6_80-6_85, or start angle identification ratio, parameter 6_58, is too high for the motor.	Check brake operation. Check motor parameters. Check motor start angle identification current.	Rotor movement is measured during start angle identification. Fault is created if the rotor moves more than two mechanical degrees during identification process.	Stops immediately.	Automatically when correct angle is found.	Increase the value of the parameter 6_58 until the rotor starts to slip during angle identification, or reduce brake force.
		26	0	Motor control fault	Motor bridge has stopped modulation during run.	Motor bridge might be broken or drive software is corrupted.	Update drive software. If problem persists then replace the drive (KR6 or KR7) or INU (KR8).	Drive internal supervision.	Stops immediately.	Automatically.	
		37	0	Device changed (same type)	Drive I/O board changed with a board that was previously installed to a drive of same type.	Drive I/O board changed from one KDM40 to another KDM40, from KDM90 to KDM90 or KDM150 to KDM150.	Update drive software.	Drive supervises software versions and IDs between the installed boards.	Stops immediately.	Automatically but drive software should be updated to latest version.	Replace drive I/O board from another unit of same type.
		40	0	Device unknown	Drive I/O board connected to a drive with unknown power board.	Drive power board does not have configuration data for line and motor bridges, or drive software is corrupted.	If problem persists then replace the drive (KR6 or KR7) or INU and AFE (KR8).	Drive supervises software versions and IDs between the installed boards.	Drive is faulted.	Replace drive.	

		41	0	Motor bridge IGBT temp	Motor bridge temperature is too high.	Motor brakes do not open, excessive load on drive, drive fan is not operating, ambient temperature is too high, switching frequency of motor bridge's transistors is too high, encoder pulse per revolution 6_64 is wrong or encoder is broken.	Check brake operation. Check drive fan (KR6 or KR7) or INU fan (KR8) operation. Check drive heatsink (KR6 or KR7) or INU heatsink (KR8) airflow. Check shaft or machine room temperature. Check switching frequency (parameter 6_66). Check encoder pulses per revolution (parameter 6_64). Check encoder operation.	Temperature of motor bridge's power transistors is estimated. Supervision trips if the temperature is higher than 135C	Stops immediately.	Automatically when fault disappears.	Disconnect drive fan, and run the elevator.
		43	0	Motor encoder fault, channel A	Encoder channel A is missing.	Encoder wire A+ and/or A- is disconnected or grounded by some reason, for example, wrong wiring in the encoder cable, short-circut between encoder wires and/or ground, encoder is broken or drive's I/O board is broken.	Check encoder wiring. Check encoder operation.	Drive supervises encoder channel's polarity. Supervision trips if both lines are simultaneously at the same state.	Stops immediately and drive is faulted.	Automatically when fault disappears.	Disconnect one of the two wires used for the encoder channel A.
		44	0	Motor encoder fault, channel B	Encoder channel B is missing.	Encoder wire B+ and/or B- is disconnected or grounded by some reason, for example, wrong wiring in the encoder cable, short-circut between encoder wires and/or ground, encoder is broken or drive's I/O board is broken.	Check encoder wiring. Check encoder operation.	Drive supervises encoder channel's polarity. Supervision trips if both lines are simultaneously at the same state.	Stops immediately and drive is faulted.	Automatically when fault disappears.	Disconnect one of the two wires used for the encoder channel B.
		45	0	Motor encoder fault, channel A and B	Both encoder channels are missing	Encoder wire B+ and/or B- and A+ and/or A- are disconnected or grounded by some reason, for example, wrong wiring in the encoder cable, short-circut between encoder wires and/or ground, encoder is broken or drive's I/O board is broken.	Check encoder wiring. Check encoder operation. Check encoder power.	Drive supervises encoder channel's polarity. Supervision trips if both lines are simultaneously at the same state.	Stops immediately and drive is faulted.	Automatically when fault disappears.	Disconnect one of the two wires used for the encoder channel B.
		48	0	Estimated speed difference	Difference limit between measured motor speed and estimated motor speed exceeded.	Wrong motor parameters in group 6_80-6_85, elevator does not follow the speed reference, speed controller gain 6_1 is too low, brakes do not open properly, there is something wrong in the motor encoder. Motor rotates to wrong direction	Check motor parameters. Check P-factor (parameter 1). Check brake control and brake operation. Check motor encoder. Check motor polarity and encoder polarity.	Drive calculates the difference between the measured motor speed (encoder) and the estimated motor speed (mathematical motor model). Supervision trips if the absolute value of the difference exceeds 0.30 m/s.	Stops immediately.	Automatically when fault disappears.	Stop motor encoder from rotating by removing it from the motor, or use external encoder, and try to run the elevator.
		50	0	Device changed (different type)	Drive I/O board changed with a board that was previously installed to a drive of different type.	Drive I/O board changed from one KDM40 to another KDM40, from KDM90 to KDM90.	Cycle power.	Drive supervises software versions and IDs between the installed boards.	Stops immediately.	By user: update drive software.	Replace drive I/O board from another unit of different type.
		51	0	Device added (different type)	Drive I/O board changed with a board that has not been in any drive before.	Drive I/O board replaced with a new one.	Cycle power.	Drive supervises software versions and IDs between the installed boards.	Stops immediately.	By user: update drive software.	Replace drive I/O board with a new one.
		52	0	DC-link overvoltage	Measured DC-link voltage is too high.	Braking chopper is used but braking resistors are not connected, high overvoltage spikes detected in the supply network, line bridge does not function properly, supply network has disappeared when elevator has moved to light direction.	Check DC link voltage from drive signal monitoring (110). Check power supply lines. Check elevator features (BMV, EBD, EPD). Check elevator fuses L1, L2, and L3. Check other fault codes.	Voltage over DC-link capacitors is measured. Supervision trips if the voltage is higher than 910 V.	Stops immediately.	Automatically when fault disappears.	Disable line bridge control, and try to run the elevator to light direction.
		54	0	Motor bridge earth fault	Earth fault in motor bridge, motor cabling or motor winding.	Wrong connection in motor cabling, broken motor cable, motor insulation fault, drive broken.	Check motor cable. If problem persist check motor.	Drive's internal supervision trips if current measurement detects that the sum of the motor currents deviates too much from zero value.	Stops immediately.	Automatically when fault disappears.	Connect one motor phase to ground via few hunder kohm resistor and try to run.
Main code	**Sub code**	**Data 1**	**Data 2**	**Name**	**Description**	**Reason**	**Action**	**Detection**	**Operation**	**Recovery**	**Testing**
		61	0	Motor bridge overtemp	Motor bridge temperature is too high.	Excessive load on drive, drive fan is not operating, ambient temperature is too high, switching frequency of motor bridge's transistors is too high.	Check drive fan (KR6 or KR7) or INU fan (KR8) operation. Check drive heatsink (KR6 or KR7) or INU heatsink (KR8) airflow. Check shaft or machine room temperature. Check switching frequency (6_66) Check drive signal monitoring for drive heatsink temperature (6_75_40).	Power unit's board temperature is measured. Supervision trips if the temperature is above 100 C.	Stops immediately.	Automatically when heatsink temperature is low enough.	Heat heatsink.
		64	0	Line bridge earth fault	Earth fault in drive, motor bridge, motor cabling or motor winding.	Wrong connection in motor cabling, broken motor cable, motor insulation fault, drive broken.	Check motor cable. If problem persist check motor.	Drive's internal supervision trips if current measurement detects that the sum of the line currents deviates too much from zero value.	Stops immediately.	Automatically when fault disappears.	Connect one motor phase to ground via few hunder kohm resistor and try to run.
		72	0	Drive system	Communication between I/O board and motor bridge control processr (INU) has failed.	Drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR) or INU (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software and if problem persists replace drive.	
		74	0	Drive system	Watchdog has reset CPU.	Drive software might be corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR) or INU and AFE (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software and if problem persists replace drive.	
		75	0	Drive system	Voltage of auxilary power in motor bridge control (INU) unit is too low.	Earth fault in drive's auxilary power supply, there might have been a black out in the supply network, drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR) or INU (KR8).	Drive internal supervision trips if -15V:Trip level=-12V +15V:Trip level=11,6V +17V:Trip level=14,2V	Stops immediately.	Automatically when fault disappears.	Connect small resistor between 24 V power supply and ground..
		76	0	Drive system	Motor bridge phase fault (INU): voltage of a output phase does not follow the reference.	Drive software is corrupted or drive is broken. Motor bridge phase fault (INU): voltage of a output phase does not follow the reference.	Update drive software. If problem persists then replace the drive (KR6 or KR) or INU (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software and if problem persists replace drive.	
		77	0	Drive system	Motor bridge (INU) CPLD has gone in fault but there is no detailed information about the fault.	Drive software is corrupted or drive is broken.	If problem persists then replace the drive (KR6 or KR) or INU (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software and if problem persists replace drive.	
		78	0	Drive system	I/O board's and motor bridge's control processor's (INU) softwares are incompatible.	Drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR) or INU (KR8).	Drive internal supervision.	Stops immediately and drive is faulted.	By user: update drive software and if problem persists replace drive.	
		79	0	Drive system	Software version cannot be read (motor bridge control processor (INU) has no software at all).	Drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR) or INU (KR8).	Drive internal supervision.	Stops immediately and drive is faulted.	By user: update drive software and if problem persists replace drive.	
		80	0	Drive system	I/O board control CPU overload.	Drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR) or INU and AFE (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software and if problem persists replace drive.	
		81	0	Drive system	Memory access has failed.	Drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR) or INU and AFE (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software and if problem persists replace drive.	
		82	0	Drive system	Necessary power-unit's device properties cannot be read.	Drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR) or INU and AFE (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software and if problem persists replace drive.	
		90	0	Drive system	Communication between line and motor bridge's control processors, AFE and INU, has failed.	Drive software is corrupted or drive is broken.	Update drive software. Check cables between AFE and INU (KR8). If problem persists then replace the drive (KR6 or KR7) or INU and AFE (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software and if problem persists replace drive.	
		92	0	Drive system	Voltage of auxiliary power in line bridge control (AFE) unit is too low.	Earth fault in drive's auxilary power supply, there might have been a black out in the supply network, drive software is corrupted or drive is broken.	If problem persists then replace the drive (KR6 or KR7) or AFE (KR8).	Drive internal supervision trips if -15V:Trip level=-12V +15V:Trip level=11,6V +17V:Trip level=14,2V	Stops immediately.	Automatically when fault disappears.	Connect small resistor between 24 V power supply and ground..
		93	0	Drive system	Line bridge phase fault (AFE): input phase voltage does not follow the reference.	There might have been a black out in the supply network, drive software is corrupted or drive is broken.	If problem persists then replace the drive (KR6 or KR7) or AFE (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software and if problem persists replace drive.	

Main code	Sub code	Data 1	Data 2	Name	Description	Reason	Action	Detection	Operation	Recovery	Testing
		94	0	Drive system	Line bridge (AFE) CPLD has gone in fault but there is no detailed information about the fault.	Drive software is corrupted or drive is broken.	If problem persists then replace the drive (KR6 or KR7) or AFE (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software and if problem persists replace drive.	
		95	0	Drive system	I/O board's and line bridge's control processor's (AFE) softwares are incompatible.	Drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR7) or AFE (KR8).	Drive internal supervision.	Stops immediately and drive is faulted.	By user: update drive software and if problem persists replace drive.	
		96	0	Drive system	Software version cannot be read (line bridge control processor (AFE) has no software at all).	Drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR7) or AFE (KR8).	Drive internal supervision.	Stops immediately and drive is faulted.	By user: update drive software and if problem persists replace drive.	
		97	0	Drive system	Cannot start some part of drive software.	Drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR7) or INU and AFE (KR8).	Drive internal supervision.	Stops immediately and drive is faulted.	By user: update drive software and if problem persists replace drive.	
		98	0	Drive system	Invalid function block used in drive's application software.	Drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive.	Drive internal supervision.	Stops immediately and drive is faulted.	By user: update drive software and if problem persists replace drive.	
		100	0	Drive updates internal software	The drive has its internal software update on-going.	Drive updates software to line and motor bridge's control processors.	Update drive software. If problem persists then replace the drive (KR6 or KR7) or INU and AFE (KR8).	Drive internal supervision.	Drive is faulted.	Automatically when software update is complete.	Update drive software with a version different from that currently shown in menu 6_97 + UP.
		101	0	Line bridge overcurrent	Measured line current is too high.	Blackout in supply network during regeneration, software might be corrupted or drive is broken.	Update drive software. Check line voltage. If problem persists then replace the drive (KR6 or KR7) or AFE (KR8).	Line current is supervised by a hardware sensors. Supervision trips if overcurrent limit is exceeded.	Stops immediately.	Automatically when fault disappears.	Short-circuit DC-link output to ground, bypass circuit breaker in the charging circuit, and power-up system.
		102	0	Dynamic brake failure	Dynamic braking feature is broken	Dynamic braking feature is broken in drive.	If problem persists then replace the drive (KR6 or KR7) or INU (KR8).	Drive internal supervision	Stops immediately.	Replace drive.	
		104	0	Drive system	Signal comparision fault	PWM enable IO, power and main contactor signal comparison failed (750ms).	Check K201, K201:3 and K201:4 wiring. If problem persists then replace the drive.	Drive internal supervision.	Stops immediately.	Replace drive.	
		105	0	Drive system	Active zero current failure	Active zero current supervision has detected current flow to motor during stop state.	Cycle power. If problem persists then update drive software. If problem persist then replace the drive (KR6 or KR7) or INU (KR8).	Drive internal supervision.	Stops immediately.	Check that motor cables are connected properly.	
		107	0	Line bridge saturation	Voltage/current generated by the line bridge does not follow the reference.	Line bridge is broken.	If problem persist then replace the drive (KR6 or KR7). Or AFE (KR8)	Drive internal supervision.	Stops immediately.	Replace drive.	
		113	0	Line bridge undertemp	Heatsink temperature is too low.	Drive temperature is below operational limits.	Check machine room temperature.	Heatsink temperature is measured. Supervision trips if the temperature is below -10 C.	Stops immediately.	Automatically when heatsink temperature is high enough.	Cool heatsink.
		114	0	Line bridge overtemp	Heatsink temperature is too high.	Drive temperature is above operational limits, excessive load on drive, drive fan is not operating, ambient temperature is too high, switching frequency of motor bridge's transistors is too high, the short-circuit impedance of the supply network is out of limits.	Check drive fan (KR6 or KR7) or AFE fan (KR8) operation. Check drive heatsink (KR6 or KR7) or AFE heatsink (KR8) airflow. Check shaft or machine room temperature. Check RT display for drive heatsink temperature (6_75_40).	Heatsink temperature is measured. Supervision trips if the temperature is above 100 C.	Stops immediately.	Automatically when fault disappears.	Heat heatsink.
Main code	Sub code	Data 1	Data 2	Name	Description	Reason	Action	Detection	Operation	Recovery	Testing
		141	0	Line bridge IGBT temp	Line bridge temperature is too high.	Drive temperature is above operational limits, excessive load on drive, drive fan is not operating, ambient temperature is too high, switching frequency of motor bridge's transistors is too high, the short-circuit impedance of the supply network is out of limits.	Check drive fan (KR6 or KR7) or AFE fan (KR8) operation. Check drive heatsink (KR6 or KR7) or AFE heatsink (KR8) airflow. Check shaft or machine room temperature. Reduce elevator speed.	Temperature of line bridge's power transistors is estimated. Supervision trips if the temperature is higher than 135 C.	Stops immediately.	Automatically when fault disappears.	
		203	0	Drive interface communication lost fault	Communication is lost between drive and DCBG.	Communication is lost between DCBG and Drive IO board.	Check cable and connectors between Drive IO board X6 and DCBG XRS1	Drive monitors RS485 communication between itself and DCBG. Supervision is tripped if a breakdown is detected.	Stops immediately and drive is faulted	Automatically when communication works.	Disconnect RS485 communication cable from connector X6 or XRS1.
		207	0	PWM Enable Missing At Start	Safety circuit not closing after 10s.	Problem in wiring between I/O board X4-connector and auxilary contactors, or both auxilary contactors are stuck open.	Check X4 connector and wiring. Check auxiary contacts of 201:3 and 201:4	Drive supervises the state of PWM ENABLE signal connected to the connector X4 in I/O board.	Stops immediately.	Automatically when fault disappears.	Stuck both auxilary contactors to open state.
		208	0	PWM Enable Missing At Stop	Safety circuit not opening after 10s.	Short-circuit in wiring between I/O board X4-connector and auxilary contactors, or both auxilary contactors are stuck closed.	Check X4 connector and wiring. Check auxiary contacts of 201:3 and 201:4	Drive supervises the state of PWM ENABLE signal connected to the connector X4 in I/O board.	Stops immediately.	Automatically when fault disappears.	Stuck both auxilary contactors to closed state.
		209	0	Main Contactor Closing	Main contactor is not closing in 10s after run commanded on.	Problem (loose contact) in wiring between I/O board X4-connector and main contactor, or main contactor is stuck open.	Check X4 connector and wiring. Check auxiary contacts of 201:1	Drive supervises the state of MAIN CONTACTOR signal connected to the connector X4 in I/O board.	Stops immediately.	Automatically when fault disappears.	Stuck main contactor to open state.
		210	0	Main Contactor not Opening	Main contactor is not opening after run commanded off in 10s.	Problem (short-circuit) in wiring between I/O board X4-connector and main contactor, or main contactor is stuck open.	Check X4 connector and wiring. Check auxiary contacts of 201:1	Drive supervises the state of MAIN CONTACTOR signal connected to the connector X4 in I/O board.	Stops immediately.	Automatically when fault disappears.	Stuck main contactor to close state.
		211	0	PWM signal fault	Hardware and software signals are different for the PWM ENABLE chain.	Drive software might be corrupted or the drive is broken.	Cycle power. Update drive software. If problem persists then replace drive (KR6 or KR7) or INU (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software and if problem persists replace drive.	Cut the wire for PWM ENABLE signal inside the drive between the I/O board and power board.
		212	0	Phase missing	One input phase is missing from the drive.	Main fuse is blown, one input phase is missing, problem in the charging circuit, or the drive is broken.	Check DC link voltage from drive real time display (6_75_110). Check power supply lines. Check elevator features (BMV, EBD, EPD). Check elevator fuses L1, L2, and L3. Check other fault codes.	Drive internal supervision.	Stops immediately.	Automatically when fault disappears.	
		213	0	Line bridge syncronization failure.	Line bridge cannot synchronize to the supply network.	Drive software might be corrupted or the line bridge is broken or supply is unstabile.	Check DC link voltage from drive real time display (6_75_110). Check power supply lines. Check elevator features (BMV, EBD, EPD). Check elevator fuses L1, L2, and L3. Check other fault codes.	Drive internal supervision.	Stops immediately.	Automatically when fault disappears.	
		214	0	Line bridge not starting	Line bridge is not starting after run command.	Drive software might be corrupted or the line bridge is broken.	Check DC link voltage from drive real time display (6_75_110). Check power supply lines. Check elevator features (BMV, EBD, EPD). Check elevator fuses L1, L2, and L3. Check other fault codes.	Drive internal supervision.	Stops immediately.	Automatically when fault disappears.	
		215	0	Invalid supply frequency	Nominal frequency of the supply network is out of limits.	Nominal frequency of the supply network is out of limits, drive software might be corrupted or the line bridge is broken.	Check DC link voltage from drive real time display (6_75_110). Check power supply lines. Check elevator features (BMV, EBD, EPD). Check elevator fuses L1, L2, and L3. Check other fault codes.	Drive internal supervision.	Stops immediately.	Automatically when fault disappears.	
		218	0	PWM/Contactor signal missing	Signals dropping during run.	Main contactor or auxilary contactor drops during run before DCBG has indicated this situation to the drive unit.	Check X4 connector and wiring. Check auxiary contacts of 201:1 and 201:4 and 201:5	Drive internal supervision.	Stops immediately.	Automatically when fault disappears.	Disconnect X4 connetor from I/O board during run.
		219	0	ZeroCurrent fault	Motor current detected although motor bridge power transistors are switched off.	Broken motor bridge. Broken current sensors.	Cycle power. If problem persists then replace drive (KR6 or KR7) or INU (KR8).	Motor current is measured by current sensors.	Drive is faulted.	By user: switch to inspection/RDF mode or cycle power.	Use external current loop and power supply to feed current through motor bridge's current sensors during run.

Main code	Sub code	Data 1	Data 2	Name	Description	Reason	Action	Detection	Operation	Recovery	Testing
		221	0	Drive Temperature	Drive is too hot	Drive fan is broken, too many starts per hour, drive is too small for the elevator application	Check drive fan (KR6 or KR7) or AFE fan or INU FAN (KR8) operation. Check drive heatsink (KR6 or KR7) or AFE heatsink or INU heatsink (KR8) airflow. Check shaft or machine room temperature. Check RT display for drive heatsink temperature (6_75_40).	Drive internal supervision indicates that heatsink temperature is higher than 90 degress Celcius.	Drive is faulted.	Automatically	
		222	0	Sequential IGBT temp	Too many IGBT temp faults (130_5000_41 or 130_5000_141) within one minute.	Drive fan is broken, too many starts per hour, drive is too small for the elevator application, motor brakes are not opening."	Check brake control and brake operation. Check motor encoder. Check motor polarity and encoder polarity. Check motor bridgre switching frequency (6_66)	Drive internal supervision indicates that drive has faulted to faults 130_5000_41 or 130_5000_141 more than 3 times during one minute	Drive is faulted.	Automatically	
		223	0	LCL overtemperature	Hardware triggered overtemperature supervision of KR8.	The temperature switch cable between LCL and I/O board is broken; the temperature switch cable connectors are not connected properly; there is not a correct flow of cooling air (LCL fan not running or filter is not clean). Ambient temperature too high.	Check LCL fan (KR8) operation. Check IO-board X3 connector and cable to LCL filter (KR8).	Hardware triggered overtemperature supervision of KR8.	Drive is faulted.	Automatically	
		224	0	Dynamic braking activated when motor is still rotating.	Drive detected that when dynamic braking is switched on rotor frequency is too high	Problem with mechanical brakes, drive software is corrupted or drive is broken.	Check deceleration during quick stop. (6_72_24). Check brake controllers and brake adjustment.	Drive internal supervision indicates that measured rotor RPM is 5% or more of nominal motor RPM when dynamic braking is switched on.	Drive is faulted.	Automatically	
	5100	16	0	Drive thermal warning	Motor is overheated according to software temperature model	Motor is too small for the elevator duty cycle, too low temperature time constant used in motor model.	Check motor cooling. Adjust motor thermal protection (70).	Drive internal supervision.	Stops immediately and drive is faulted.	Automatically when motor temperature is low enough.	Decrease temperature time constant and run motor for a while.
		33	0	Drive fan speed warning	Fan speed does not follow the speed reference totally but it is still possible to run the drive	Drive fan speed does not follow reference.	Check drive fan (KR6 or KR7) or INU and AFE fan (KR8). Check drive fan lifetime counters 1 (KR6 or KR7) or 1 and 2 (KR8).	Drive internal supervision.	Warning is displayed.	Automatically when fan follows the reference.	
		34	0	Drive fan 1 life-time exceeded	Drive fan has been running longer than 100% of it's expected life-time.	Drive fan life-time exceeded.	Check drive fan lifetime counters 1 (KR6 or KR7 or KR8). Replace drive fan (KR6 or KR7) or INU fan (KR8).	Drive internal supervision.	Warning is displayed.	By user: reset drive fan's life-time counter, and replace drive fan with a new one.	
		134	0	Drive fan 2 life-time exceeded	Drive fan 2 has been running longer than 100% of it's expected life-time.	Drive fan life-time exceeded.	Check drive fan lifetime counters 2 (KR8). Replace AFE fan (KR8).	Drive internal supervision.	Warning is displayed.	By user: reset drive fan's life-time counter, and replace drive fan with a new one.	
Main code	**Sub code**	**Data 1**	**Data 2**	**Name**	**Description**	**Reason**	**Action**	**Detection**	**Operation**	**Recovery**	**Testing**
		42	0	Encoder acceleration warning	Measured encoder signal indicates accelerations higher than it is physically possible to achieve.	Contact surface between motor encoder wheel and motor brake wheel has bumps or dirt on it, encoder is not fixed properly, encoder is broken internally: e.g. pulses are missing, or encoder signals are disturbed by electromagnetic interference (EMI).	Check encoder wheel. Check encoder fixing. Check encoder wiring. Check encoder operation.	Drive calculates elevator acceleration from the speed measured by the motor encoder. Supervision trips if the absolute value of the calculated acceleration is higher than 20 m/s2.	Warning is displayed.	Automatically	Add adhesive tape to one part of the motor brake wheel, use encoder wheel with non-uniform diameter, or use encoder with a specified defect, and run the elevator.
		49	0	Encoder wear warning	Motor encoder produces different amount of pulses per one motor revolution than it is expected.	Wrong values in the parameters: 6_3, 6_7, 6_63, 6_64, 6_80-6_85, encoder wheel is worn, encoder has different properties than is assumed, wrong encoder.	Check parameters (3, 7, 63, 64, 80-85). Check encoder operation. Check encoder fixing. Check Motor encoder estimator (29).	Drive monitors the amount of encoder pulses detected per one motor revolution against a value defined by parameter 6_64. The monitored value is an estimate taken as the average of the values detected when the elevator runs to both up and down directions at the defined nominal speed 6_1. If the monitored value differs more than 0.5 % but less than 2.0 % from the pre-defined value, motor control starts to use the new value. If the new value differs more than 2.0 % from the pre-defined value, a warning is produced.	Warning is displayed.	Automatically	Set signal 29 to the real-time display 6_75. Set parameter 6_64 few percents away from the nominal. Run the elevator to up and down directions at full speed, and check the value shown in the real-time display.
		59	0	Motor bridge overtemp warning	Heatsink temperatue is above 90 C .	Drive fan is not rotating fast enough.	Check drive fan (KR6 or KR7) or INU fan (KR8) operation. Check drive heatsink (KR6 or KR7) or INU heatsink (KR8) airflow. Check shaft or machine room temperature. Check switching frequency (6_66) Check drive signal monitoring for drive heatsink temperature (6_75_40).	Heatsink temperature is measured. Supervision trips if the temperature is above 90 C.	Drive is faulted at floor.	Automatically when heatsink temperature has been decreased enough.	Disconnect drive fan, and run the elevator.
		73	0	Drive system	Communication between I/O board and motor bridge control processor (INU) has interference but is still working.	Drive software is corrupted or drive is broken.	Update drive software.	Drive internal supervision.	Stops immediately.	By user: update drive software or replace drive.	
		91	0	Drive system	Communication between line and motor bridge's control processors, AFE and INU, has interference but it is still working.	Drive software is corrupted or drive is broken.	Update drive software. If problem persists then replace the drive (KR6 or KR7). Check cables between AFE and INU (KR8).	Drive internal supervision.	Stops immediately.	By user: update drive software or replace drive.	
		99	0	CPU resource overload warning.	CPU load is too high.	Drive software might be corrupted.	If problem persists then replace the drive (KR6 or KR7) or INU and AFE (KR8).	Drive internal supervision.	Warning is displayed.	Automatically	
		103	0	Drive system	Drive memory is corrupted	Drive power board XML data is corrupted. Drive is still usable if I/O board is not changed.	If problem persists then replace the drive (KR6 or KR7) or INU and AFE (KR8).	Drive internal supervision.	Warning is displayed.	Automatically. Drive should be replaced.	
		220	0	Motor bridge overcurrent warning	Measured motor current is at the maximum limit drive can produce.	Brakes do not open, mechanical problem hindering movement, wrong motor parameters in groups 6_64, 6_65, 6_80-6_85, broken encoder, short-circuit or earth fault in motor cables or broken motor current sensor.	Check elevator parameters (6_1, ..., 6_14). Check machinery parameters (6_60, ..., 6_65 and 6_80, ..., 6_91) Check brakes. Check balancing. Check shaft mechanics. Check motor wiring. Check motor connectors. Check drive connectors.	Motor current is measured. Warning is displayed when current is limited to drive's maximum output.	Warning is displayed.	Automatically	Disconnect brake cable from brake control module and try to run elevator.
		225	0	IGBT Temp (INU)	INU IGBT transistor or diode temperature too high	INU IGBT Temperature too high.	Check acceleration. Check balancing. Drive may be too small for this elevator. Check switching frequency (parameter 6_66).	Triggered if maximum calculated INU IGBT transistor or diode temperature differs from heatsink temperature more than the allowed limit.	Warning is displayed.	Automatically	
		226	0	IGBT Temp (AFE)	AFE IGBT transistor or diode temperature too high	AFE IGBT temperature too high.	Check acceleration. Check balancing. Drive may be too small for this elevator. Check switching frequency (parameter 6_66).	Triggered if maximum calculated AFE IGBT transistor or diode temperature differs from heatsink temperature more than the allowed limit.	Warning is displayed.	Automatically	
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[redacted]	1061	0	0	NTS processor interface communication lost fault	Communication is lost between DCBG main CPU and NTS CPU	DCBG board is broken , update process failed or SD card was removed before update process was completed.	Update DCBG software. If problem persist replace DCBG board.	DCBG main CPU monitors communication between itself and NTS processor. Supervision is tripped if a breakdown is detected.	Drive is faulted, this can happen only after sw update process.	Automatically when software is re-programmed again.	
	4000	0	0	Drive interface communication lost fault	Communication is lost between drive and DCBG.	Problem with drive communication, intermittent power supply break if with code 0170, 0026.	Update software for DCBG and drive software. Check cable and connectors between Drive IO board X6 and DCBG XRS1 Cycle power. If problem persist replace DCBG and /or IO-board board.	DCBG monitors RS485 communication between itself and DCBG. Supervision is tripped if a breakdown is detected.	Stops immediately and drive is faulted	Automatically when communication works.	Disconnect RS485 communication cable from connector X6 or XRS1.

Drive LEDs

						Operation

Location	Name	Position	Color	Description		On	Off	Blinking
DCBG	EC_MCU	LED201	Yellow	MCU processor status. KDM Drive SW in SD card.		If SD card has been inserted and KDM Drive SW package has been found into te SD card.	MCU processor is ok.	DCBG draws power from XBAT1 instead of XAP1.
DCBG	SD_CARD	LED203	Green	SD card operation.		SD card has been read.	No SD card detected.	SD card is being read.
DCBG	EC_NTS	LED301	Yellow	NTS processor status.		NTS function is active.	NTS function is inactive.	1) MCU processor has activated NTS function, 2) NTS function has been completed.
DCBG	MC INPUT	LED401	Yellow	Main contactor status.		Main contactor picked.	Main contactor released.	Not defined.
DCBG	POWER	LED701	Green	+24V supply is OK.		24V supply is OK.	24V supply is missing or too low.	Not defined.
DCBG	CHANGE BOARD	LED702	Red	Software is running.		Set on by hardware when board is powered.	Software is running.	Not defined.
I/O board	FLT	H1	Red	Drive is in fault.		Software or hardware triggered fault is active in the drive.	Drive is OK.	Drive software is being updated.
I/O board	>50VDC	H11	Yellow	Dangerous voltage in DC-link capacitors.		Measured DC-link voltage is above 50 V.	Measured DC-link voltage is below 50 V.	Not defined.
I/O board	COM	H14	Green	Run state.		Motor bridge is modulating.	Motor bridge is not modulating.	Not defined.
MC board	AFE STATUS	AFE	Green	Line bridge control processor status.		Line bridge is active.	Line bridge is inactive.	1) Processor tries to reset itself, 2) Software update in progress, 3) Communication is broken.
MC board	INU STATUS	INU	Green	Motor bridge control processor status.		Communication is OK between processors inside the drive.	MC board is broken.	1) Processor tries to reset itself, 2) Software update in progress,

End of table.